Changing the Game

North America

Largest Market

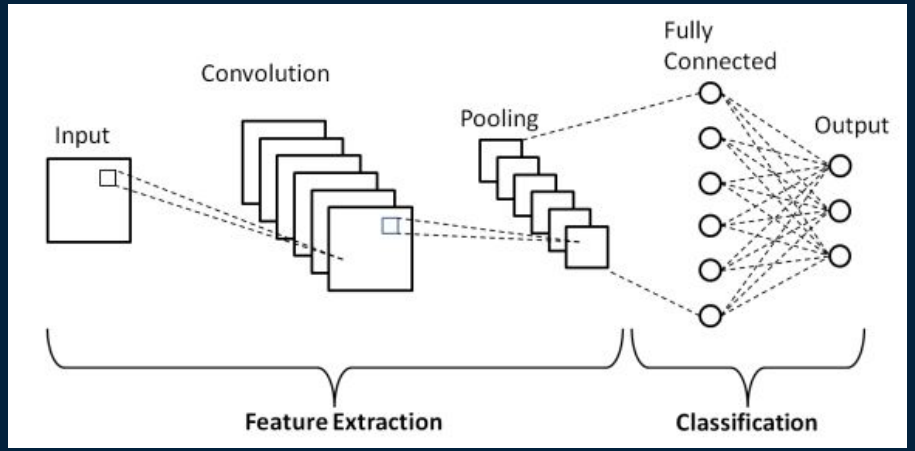
$61.3B

Global Market Value

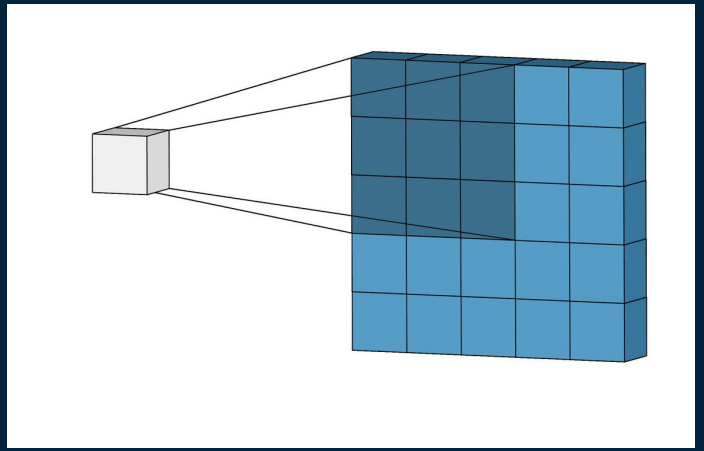
14.6 %

Annual Growth Rate

What is a Convolutional Neural Network?



(researchgate.net)



(towardsdatascience.com)

Surrounding Literature

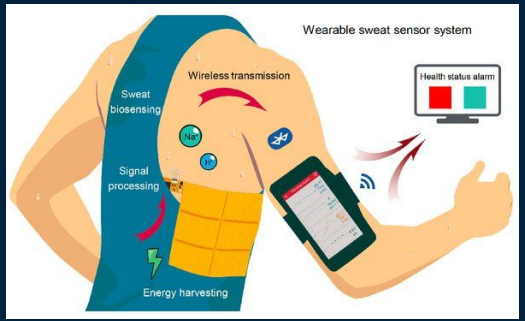
Literature Review

Li et al.

- Monitor real-time physiologic and movement parameters
- Screen for potential injuries
- Increased demand



(techcrunch.com)

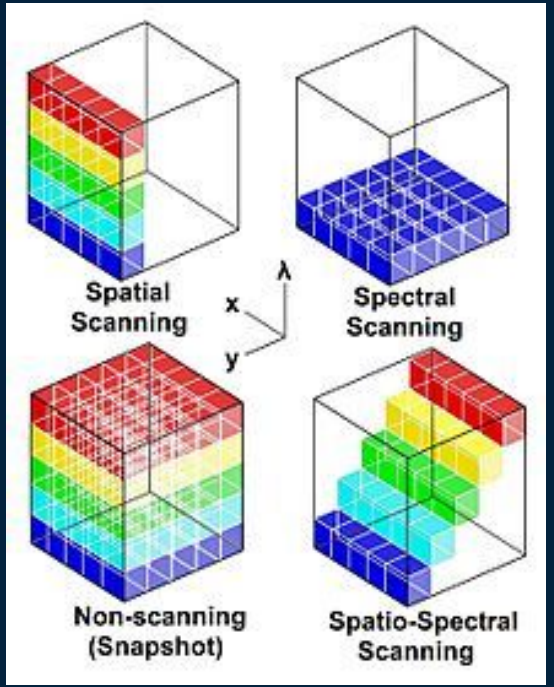


(caltech.edu)

Literature Review

Hasan et al.

- Hyperspectral image classification
- CNN vs. SVM
- 98.72 % Accuracy from CNN

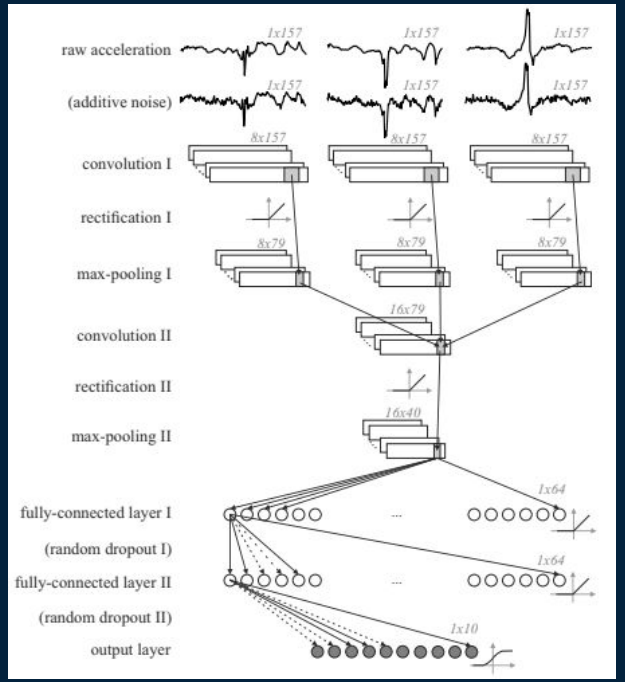


(wikipedia.com)

Literature Review

Kautz et al.

- Action classification in Volleyball
- Combined wearable devices and CNN to classify
- CNN outperformed other method by more than 16%



(Kautz et al.)

The Gap

What Is Known

- Combination can be used for numerous sports sports
- There is a demand for automation in data analytics in sports

What Is Not Known

- The use of wearable sensors and CNNs in soccer
- If it is possible to create an inexpensive wearables sensor

Research Question

How feasible is the implementation of convolutional neural networks using quantitative data collected from wearable devices in soccer action classification?

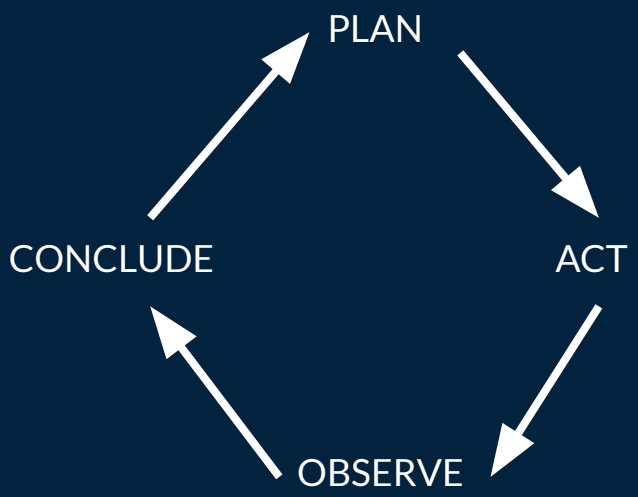
Hypothesis

Due to the flexibility of wearable sensors and CNNs, this combination will be able to classify a variety of actions in soccer.

Action Methodology

Method Justification

- No available wearable sensor
- Simultaneously investigates and solves an issue
- Hollaus et al. paper on catch detection



Method Steps



Design the Wearable Device

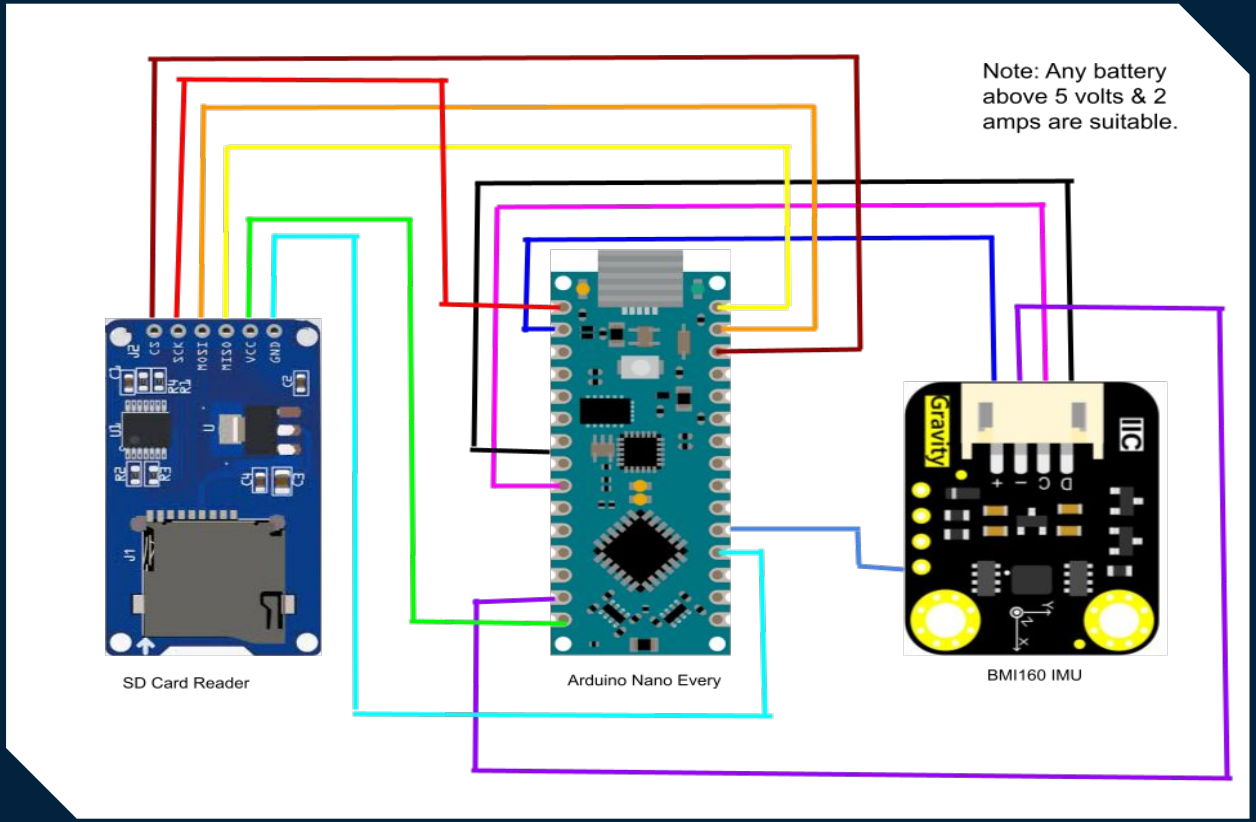


Design the CNN



Test the Combination
& Input Data Into the CNN

Wearable Sensor Design



CNN Design



Input

Conv1D + Max Pooling

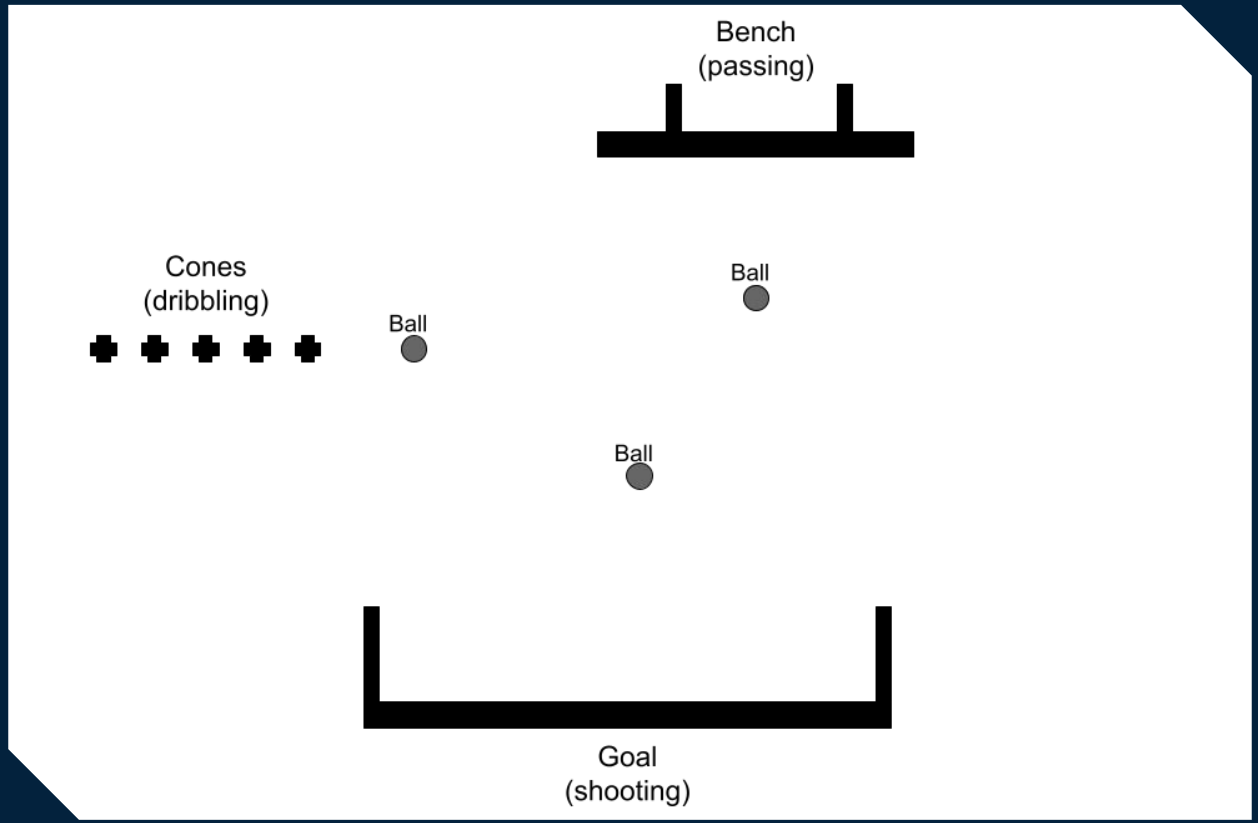
Conv1D + Max Pooling

Dense + Flatten

Output

Testing

- 3 training Sets:
  - 5 shots, passes, dribbles
- 1 test set:
  - any amount of actions/orders
- Metrics Collected:
  - Precision, Recall, F-1 Score, & Accuracy

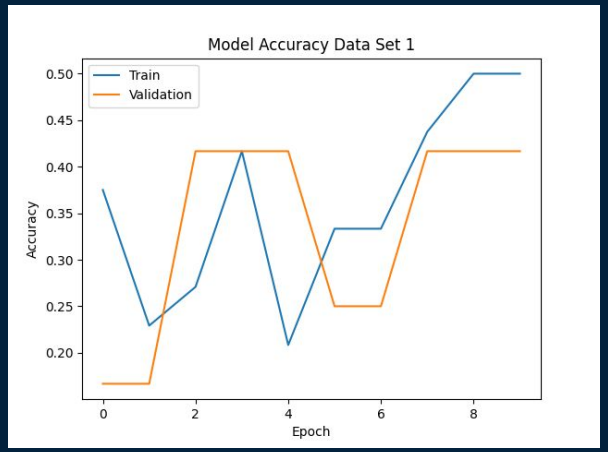

Bench
(passing)
Cones
(dribbling)
Ball
Ball
Ball
Goal
(shooting)

Results & Discussion

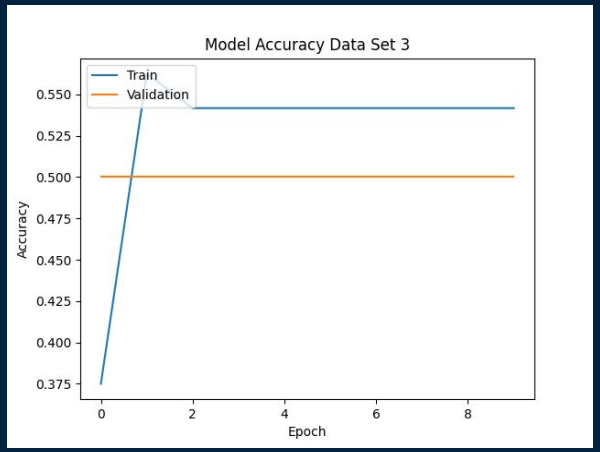
Wearable Device Results

Metric	Max Movement Acceleration (m/s^2)	Min Movement Acceleration (m/s^2)	Max Angular Acceleration (rad/s^2)	Min Angular Acceleration (rad/s^2)
Value	76.27	0.62	2.00	0.10

CNN Accuracy



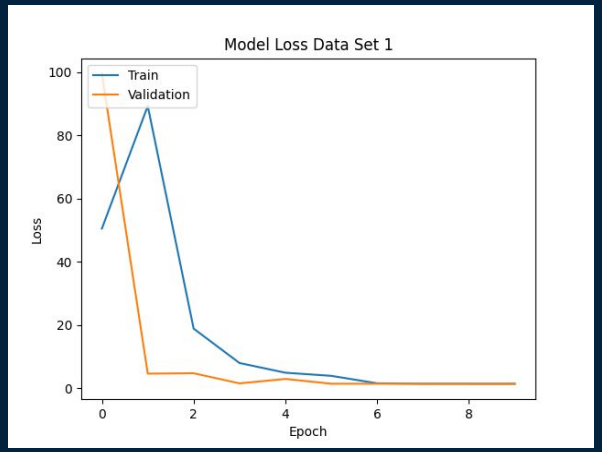


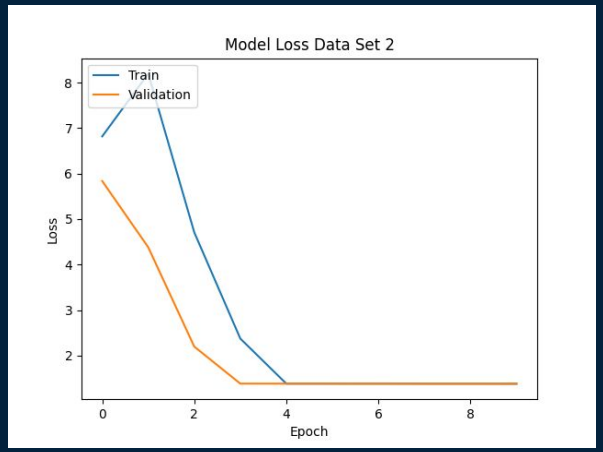


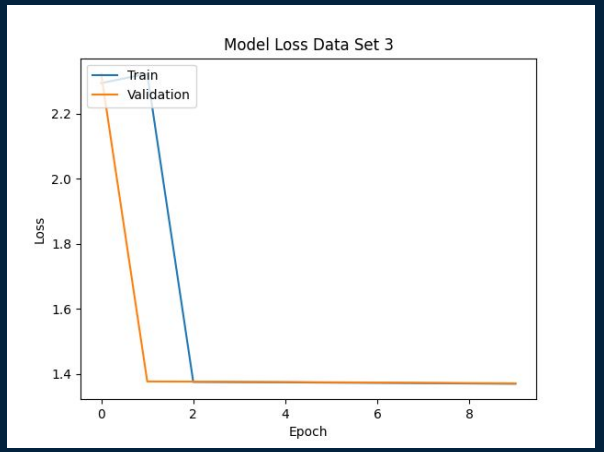
Min: 8%

Max: 55%

CNN Loss Rate







Min: 100

Max: 1.4

CNN Metrics

Metric	Precision	Recall	F1-Score	Accuracy
Value	0.0625	0.1875	0.0938	0.4667

Limitations

- Data abstraction
  - Range in comparison to Hollaus et al.
- Physical limitation
- Complexity vs. Size of CNN

Implications

- Wider spectrum of data collection
  - Accelerations, Force, microphone, etc…
- Need for other test of combinations
- More automation for soccer analytics

Significance

Despite proving the hypothesis incorrect, the results of this research points to the need of more advanced wearables sensors to collect data.

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Thank you